

Component Part List

1. **BQ79656EVM for BQ79656-Q1 - QTY: 1x**

We chose the BQ79656EVM as our primary battery monitoring platform because it provides a fully supported evaluation environment for rapid bring-up and learning. Although the newer BQ79758 device offers improvements such as improved accuracy, integrated overcurrent protection, and lower current consumption, it does not currently offer the same level of evaluation board accessibility and ecosystem support for quick prototyping.

At this stage of the project, our priority is hardware validation, communication learning, and rapid development rather than optimizing for automotive production-level features. The evaluation board allows us to experiment with real cells, configure registers via GUI, validate balancing behavior, and understand communication protocols before integrating with our MCU.

Historically, we considered the BQ79758 due to its improved protection integration, but we determined that the availability of the BQ79656EVM ecosystem provides more educational and development value for our current capstone prototype phase.

Link: [BQ79656EVM-021 Evaluation board | TI.com](#)

2. **USB2ANY Communication Interface - QTY: 1x**

The BQ79656EVM supports UART and SPI communication to a host controller. For the initial validation and learning phase, we chose to use TI's USB2ANY interface in combination with the BQ GUI rather than immediately interfacing the EVM directly with our STM32 MCU.

The primary justification for selecting USB2ANY is risk reduction and development efficiency during hardware bring-up. The USB2ANY serves as a known-good, vendor-supported USB-to-UART bridge that allows direct communication between a PC and the BQ79656EVM without requiring any firmware development. This enables us to validate cell wiring, verify voltage measurements, test balancing behavior, and confirm proper device wake-up and register configuration before introducing firmware complexity.

Through the GUI, we can directly observe how register values change when enabling balancing, setting overvoltage and undervoltage thresholds, or adjusting timing parameters. These known-good register configurations can then be replicated in C when transitioning to MCU-based control. The GUI does not permanently program the device; it simply sends runtime commands over UART, which

mirrors exactly what our MCU firmware will later implement.

We considered skipping USB2ANY and interfacing the STM32 NUCLEO-C092RC directly from the beginning. However, that approach would require immediate development of the UART protocol stack, wake-up signaling, packet framing, and error handling before confirming hardware functionality. For a first-time bring-up of a complex battery monitor IC, this would increase integration risk and slow development.

Link:  [USB2ANY Evaluation board | TI.com](#)

3. **Samsung 25R 18650 NMC Cells - QTY: 6x minimum**

We selected Samsung 25R 18650 cells because they are widely available, reliable, well-characterized, and capable of far higher current than our low-power test setup requires. Also, the previous car that VMS had used these specific cells, so we chose the same for consistency.

Although our final system concept may involve 48V modules and potentially LFP chemistry for safety, we chose NMC 18650 cells for the prototype due to cost, availability, and ease of handling.

The 6S configuration (25.2V max) is safely within the operating range of the BQ79656 and allows us to meet the minimum cell requirement (6 cells) stated in the EVM documentation.

Link:  [Samsung 25R 18650 2500mAh 20A Battery](#)

4. **Modular Single-Cell 18650 Holder for DIY Series Connection - QTY: 1x**

We deliberately chose modular single-cell holders instead of spot-welded nickel strip packs. The goal of this phase is to develop and validate the BMS, not to optimize mechanical battery pack design. Modular holders allow us to avoid spot welding, easily swap or remove cells, access every cell junction for sense wiring, modify configurations quickly, and reduce the safety risk during prototyping.

We initially explored pre-built 6-cell holders and welded packs, but those options either lacked access to individual cell taps or added unnecessary mechanical complexity for a learning platform.

Link:  [Bistook 6-Pack 18650 Battery Holder with Side Covers – Transparent Case, No Soldering, Brass Contacts, Safe & Expandable DIY Battery Storage](#)

5. 18650 Battery Cell Charger - QTY: 1x

For the initial prototyping phase, we chose to use an off-the-shelf multi-slot smart charger to charge our 18650 cells individually rather than implementing pack-level CC/CV charging through a lab supply or balance charger. This decision was made to reduce wiring complexity, minimize safety risk, and accelerate early development. A smart charger independently manages each cell's charging profile, automatically applying the correct constant-current/constant-voltage algorithm and terminating charge when the cell reaches its safe upper limit. This removes the need for manual CC/CV configuration and eliminates the possibility of operator error during early experimentation.

Because our prototype uses removable single-cell holders, charging cells individually outside the 6S configuration simplifies the workflow: cells can be safely charged, verified, and then reinstalled into the pack for testing with the BQ79656EVM. While this approach does not allow pack-level balance charging during assembly, it aligns well with our current objective of validating BMS measurement and communication functionality rather than developing a final charging architecture. We initially considered using a bench power supply or a 6S balance charger, but determined that a multi-slot smart charger provides a safer, faster, and lower-complexity solution for this early validation phase.

Link: [18650 Battery Charger, Smart Universal Charger with LCD Display for Rechargeable Batteries](#) or similar

6. 10kΩ NTC Thermistors for Temperature Monitoring - QTY: 8x minimum

The BQ79656 supports GPIO-based thermistor measurement for pack temperature monitoring. We will use standard 10k NTC thermistors because they are industry-standard for battery monitoring, offer predictable response curves, and are supported directly by the EVM bias circuitry.

PTC thermistors were considered but rejected because NTC devices are more common for temperature sensing in lithium-ion applications due to smoother, monotonic resistance curves.

Link: [https://www.amazon.com/dp/B0CH1CC1S6](https://www.amazon.com/dp/B0CH1CC1S6/ref=cm_sw_r_cso_cp_apin_dp_GD945JKWT07JSYWFH3JZ&social_share=cm_sw_r_cso_cp_apin_dp_GD945JKWT07JSYWFH3JZ&th=1)
[ref=cm_sw_r_cso_cp_apin_dp_GD945JKWT07JSYWFH3JZ&social_share=cm_sw_r_cso_cp_apin_dp_GD945JKWT07JSYWFH3JZ&th=1](https://www.amazon.com/dp/B0CH1CC1S6/ref=cm_sw_r_cso_cp_apin_dp_GD945JKWT07JSYWFH3JZ&social_share=cm_sw_r_cso_cp_apin_dp_GD945JKWT07JSYWFH3JZ&th=1) or similar

7. Small Resistor or Light Bulb as a Load - QTY: 1x

A load is not required to bring up the BQ79656EVM. We will first validate static cell measurement and balancing behavior without discharge current.

A resistive load may later be added to observe voltage drop behavior and test undervoltage thresholds under discharge conditions.

Link: No specific component

8. **Fuse for Power Protection - QTY: 1x**

Although our testing currents are low, lithium-ion cells can deliver very high fault currents if shorted. An inline fuse on Pack+ adds a simple, inexpensive layer of protection against accidental short circuits during prototyping.

We are not implementing contactors or MOSFET cutoff stages at this phase because the objective is measurement and balancing validation rather than full protection system development.

Link: No specific component

9. **STM32 NUCLEO-C092RC Host MCU - QTY: 1x**

We selected this MCU because the VCU team is using the same platform, which simplifies development, toolchains, and communication architecture.

The MCU supports UART and CAN-FD, aligning with our eventual system integration requirements.

We considered TI's TMS570 LaunchPad because it is referenced in the BQ documentation, but we determined that using our selected STM32 platform aligns better with our full system architecture and avoids unnecessary vendor lock-in.

Link:  [STMicroelectronics NUCLEO-C092RC](#)